

Report 242

Self Modified Ascent Sequences and Positive Diagonal Tables

Exact saddle expansions, dimension laws, binary probabilities and inversion

5 October 2026

Abstract

Let b_N count nonnegative upper-triangular integer tables of total sum N with positive diagonal, and let c_N count their binary members. These are the prior self-modified ascent-sequence models, with the current indexing $b_N = A098569(N-1)$ and $c_N = A121690(N-1)$ for $N \geq 1$. Starting from their established exact sums, we prove arbitrary fixed-order expansions at the exact gamma saddle, with global tail bounds and an explicit lattice transfer. Bounded real dimension marks are allowed. A full-span local Gaussian law and moment corrections determine the row distribution of A098568. If $u(u+2)e^u = 2N$, the counts have elementary relative carriers proportional to

$$\exp\left(\frac{Nu(u+1)}{u+2} \pm \frac{u}{2} \pm \frac{u^2}{4}\right),$$

with prefactor $\sqrt{2/(u^2+4u+2)}$. The binary probability is asymptotic to $e^{-u^2/2-u}$; we give its explicit rational first correction and an all-fixed-order quotient algorithm. Smooth exact-saddle inverses yield two-ceiling integer brackets, and elementary inverse seeds have vanishing absolute error. The log-square factor, the rare-event tail scale and the rounding qualifications are essential. Exact enumeration and model identifications are credited to the historical literature. Finite reproducible computations are consistency checks, not interval certificates or substitutes for the proofs.

[write] **Status and trust boundary, added 5 October 2026, batch 102.** This is bundle Report 242, filed in ProveIt as a single-source research report (provenance, credits, notation and the collected non-claims in Section 1.4). It is AI-assisted, unrefereed and not formalized; its title page and PDF metadata name no person (author “Report 242”). The exact enumeration, the word/table bridge, the binary subclass, the multiplicity substitution and the $N \log N$ scale are prior results of the literature and of Václav Kotešovec’s OEIS postings, credited in Section 1.4; the page-level readings of those sources are the source’s (**SOURCES.md**), not checked by ProveIt’s intake. All O -constants and thresholds are non-effective, and the numerical checks of Section 10 are not interval certificates. The report answers the question of OEIS A098568 on the distribution of the row terms (Section 5 and the note at its end).

1 The models and their exact enumeration

All logarithms are natural. A table of dimension m is an m -by- m upper-triangular matrix of nonnegative integers. Its size is the sum of its entries, rather than its dimension. We require every diagonal entry to be positive. The empty table has size and dimension zero and is counted once. Let $T(N, m)$ count tables of size N and dimension m , and put

$$b_0 = 1, \quad b_N = \sum_{m=1}^N T(N, m) \quad (N \geq 1). \quad (1.1)$$

Write

$$M(x) = \frac{x(x+1)}{2}, \quad q = N - x. \quad (1.2)$$

Subtracting one from every diagonal entry leaves a weak composition of $N - m$ into $M(m)$ cells. Hence the established enumeration is

$$T(N, m) = \binom{M(m) + N - m - 1}{N - m}, \quad 1 \leq m \leq N. \quad (1.3)$$

The binary subclass has all diagonal entries equal to one. Its remaining $N - m$ ones occupy distinct off-diagonal cells, so

$$U(N, m) = \binom{m(m-1)/2}{N - m}, \quad c_0 = 1, \quad c_N = \sum_{m=1}^N U(N, m). \quad (1.4)$$

An infeasible binomial coefficient in (1.4) is zero. In particular $b_1 = c_1 = 1$.

1.1 Self modified ascent sequences

An ordinary ascent sequence is a word $a_1 \cdots a_N$ of nonnegative integers satisfying $a_1 = 0$ and

$$0 \leq a_i \leq 1 + \text{asc}(a_1 \cdots a_{i-1}) \quad (i \geq 2),$$

where an ascent is a strict increase between consecutive entries. In the modification operation of Bousquet-Mélou, Claesson, Dukes and Kitaev [3], an ascent to a value v increases by one every earlier entry at least v . A sequence is self modified if this operation leaves it unchanged. Equivalently, each ascent top is a new strict maximum. Indeed, unchangedness forces precisely that condition at every processed ascent; conversely, under the condition there is no earlier entry to increase.

The resulting factorization is

$$0A_0 1A_1 \cdots (m-1)A_{m-1}, \quad (1.5)$$

where A_i is a weakly decreasing word with letters in $\{0, \dots, i\}$. The first appearances of the maxima cannot skip a value, by the ascent-sequence inequality, and each intervening segment must be weakly decreasing. Conversely every word (1.5) is such an ascent sequence and is self modified. There are $m - 1$ ascents and maximum $m - 1$.

For completeness, the explicit table bijection sends block iA_i to column $i + 1$: its entry in row $j + 1$ is the number of occurrences of j in that block. This is upper triangular and has positive diagonal. To invert, read the multiset of each column in weakly decreasing order, then concatenate the columns. Thus size is word length and dimension is one plus the maximum, or one plus the number of ascents. This bridge and its refinements are prior results [3, 4].

The binary subclass makes every block strictly decreasing, starting with its maximum. It is the $d = 1$ specialization of self-modified difference ascent sequences, whereas all positive-diagonal tables give $d = 0$; see Cerbai, Claesson and Sagan [5], Lemma 4.3 and Theorem 4.4. We use these two specializations only and make no claim here about other fixed difference parameters.

1.2 Formal series and current OEIS indexing

Marking dimension by h gives the formal identities

$$B(z, h) = 1 + \sum_{m \geq 1} \frac{h^m z^m}{(1 - z)^{m(m+1)/2}}, \quad (1.6)$$

$$C(z, h) = 1 + \sum_{m \geq 1} h^m z^m (1 + z)^{m(m-1)/2}, \quad (1.7)$$

$$B(z, h) = C\left(\frac{z}{1 - z}, h\right). \quad (1.8)$$

These are coefficientwise formal series: each fixed power of z receives finitely many summands. Equation (1.8) also follows by expanding each chosen binary cell into a positive multiplicity. It implies, for $N \geq 1$,

$$b_N = \sum_{j=1}^N \binom{N-1}{j-1} c_j. \quad (1.9)$$

Both the exact formulas and this substitution have historical antecedents [1, 2].

The current sequence conventions [10] are

$$\begin{aligned} b_N &= \text{A098569}(N - 1), \\ c_N &= \text{A121690}(N - 1) \quad (N \geq 1), \\ \text{A098568}(n, k) &= T(n + 1, k + 1) \quad (0 \leq k \leq n). \end{aligned} \quad (1.10)$$

For example, at sizes $N = 1, \dots, 9$, the values of b_N are

$$1, 2, 5, 14, 43, 143, 510, 1936, 7775.$$

The row at $N = 4$ is $(1, 6, 6, 1)$, which is row $n = 3$ of A098568. Keeping the shift in (1.10) is necessary when applying every asymptotic or inverse statement below.

1.3 Historical boundaries and the problem addressed

Bayoumi, El-Zahar and Khamis [1] already identify the exact binary interval subclass and the multiplicity substitution. Khamis [2] refines by height: Lemma 4.1 gives (1.4), and Theorem 4.2 gives (1.8). Their strictly upper-triangular block matrix has one more row than the present table; deleting its forced zero first column and last row turns its positive superdiagonal into our positive diagonal. Their height is our dimension m .

There is an important terminology distinction. These historical papers use N -freeness in the *directed covering graph*. This must not be silently replaced by induced-subposet N -avoidance or by the series-parallel/Catalan class. The positive-diagonal class in Dukes–McNamara [4] is their larger class with a chain meeting every strict-downset level. We consequently state our results in table and word language.

The 1989 asymptotic theorems concern the larger class of all covering-graph N -free posets, using the interval subclass for a lower bound; they do not give the relative formulas proved here. The leading $N \log N$ scale is prior, and finer logarithmic expressions already appear in Václav Kotešovec's OEIS postings [10]. An asymptotic equivalence for a displayed logarithmic expression, by itself,

does not establish all of its lower-order terms with additive error $o(N)$, much less a relative count equivalent. Exact enumerative antecedents also include [6, 7, 8].

Our purpose is the relative analysis of (1.3): the exact saddle carrier, arbitrary fixed correction orders, a full-lattice dimension law, a binary rare-event ratio and controlled integer inversion. The row-distribution question explicitly posed in A098568 is answered by Section 5. General Fishburn-matrix results, such as [9], concern different generating functions and do not automatically transfer to this positive-diagonal model. The source search described in the accompanying source note is bounded; no worldwide novelty, external peer review or formal proof verification is claimed.

1.4 Provenance, credits and reading conventions

[write] **Added 5 October 2026, batch 102.** *Provenance.* This report is Report 242 of the session bundle that ProveIt’s docs/incoming intake processed as batch 102: the archive **Report242.zip** (wrapper directory **Report242/**, 31 files, 638,320 bytes), arrival commit 60f54ea06, placed in 6ab1f1979. The text is the delivered **article.tex** with its twelve input files **sections/01_models.tex** to **sections/12_references.tex**, kept as delivered in that layout; the delivered 24-page PDF **Report242.pdf** and the checksum manifest **MANIFEST.sha256** are not shipped (both are retrievable from the arrival commit). The package names no ProveIt commit, path or report, so the report has no pin; its title page and PDF metadata name no person (author “Report 242”), and it carries no “prepared for private review” line. It is a single-source report, so no merge choice was made. The write prefixed the 126 delivered labels with **pdt**: (“positive-diagonal tables”; the prefix **sma**: is already used by **a124380-signed-moment-asymptotics**), added the notes marked [write], this subsection, Remarks 2.2 and 8.2, Proposition 5.3 and the bound of the note after Theorem 9.2, both with proofs, and Section 11.1. No statement, proof or number of the source was changed. Theorem numbers follow the section counter (Theorem 2.1 is Theorem 2.1, and so on).

Question answered. OEIS A098568 asks, in a comment, “How do the terms of row k tend to be distributed as k grows?” (read on 5 October 2026). With the shifts $N = n + 1$ and $D_N = K_n + 1$ of (1.10), Theorem 5.1, Proposition 5.2 and the write’s Proposition 5.3 answer it: a full-span local Gaussian law with explicit corrections, centre $r_{n+1}(0) - 1 \sim 2n/\log n$ and variance $\sim 2n/(\log n)^2$, a relative Gaussian law up to $|z| = o(n^{1/6})$, and the exact shape (5.6) beyond.

Inherited material, as the manuscript credits it (page-level readings from the delivered **SOURCES.md**; ProveIt’s intake did not consult these papers):

- the counts (1.3) and (1.4): Bayoumi, El-Zahar and Khamis [1] (Lemma 4, the binary finite sum, for a block matrix with one more row), Khamis [2] (Lemma 4.1, height-refined), Bousquet-Mélou, Claesson, Dukes and Kitaev [3] (Section 4.4, Propositions 10–11: the self-modified factorization, exact generating function and dimension-refined sum), and the further enumerations of Kube and Ruskey [6] (Theorem 1.3), Bean, Claesson and Ulfarsson [7] (Propositions 13–14, permutation models) and Chen, Fan and Zhao [8] (Theorems 1.3–1.4);
- the word/table bijection (1.5): [3] and Dukes and McNamara [4] (Lemma 3.4, with Definition 3.3, Propositions 3.6–3.7 and Corollary 3.8); the manuscript re-explains it for self-containment;
- the binary (rigid) subclass: [1] (Lemmas 2–3, the interval characterization) and [2]; as the $d = 1$ case of self-modified difference ascent sequences, Cerbai, Claesson and Sagan [5] (Lemma 4.3, Theorem 4.4);
- the multiplicity substitution (1.8) and hence the binomial transform (1.9): [1] ($I(x) = J(x/(1-x))$) and [2] (Theorem 4.2);
- the $N \log N$ scale: [1] (Theorems 5–6, logarithmic-scale estimates for the larger class of covering-graph N -free posets), and the finer logarithmic expressions posted by Václav Kotešovec in the OEIS entries A098569 and A121690 [10] (the latter dated 1 July 2025), which the inspected records state without a cited proof or a relative error. The coarse consequence (6.7) is in that line; Theorem 6.1 is the source’s relative strengthening.

Repository. Before batch 102 no file of the repository mentioned A098569, A121690, A098568, self-modified ascent sequences or positive-diagonal tables, so the source’s statement that its bounded repository comparison found no duplicate was true at placement. One batch-102 sibling meets this report: Report 243, filed as `generating-functions-and-asymptotics/oeis-sequence-asymptotics/a202059-ascent-100-110-growth` (written concurrently), bounds ascent sequences avoiding 100 by the same triangular sum, which it calls b_n with $b_n = A098569(n-1)$, and proves in its Section 2 (as delivered, Lemma 2.1) that $\log b_n = n(\log n - 2 \log \log n + \log 2 - 1) + O(n \log \log n / \log n)$. Its proof uses the comparison phase H_N of (3.1) and the same two global gaps, $4 \log 2 - 2$ and $2 - 2 \log 2$, as Lemma 3.1; its estimate follows from Theorem 6.1 on expanding $u = \log N - 2 \log \log N + \log 2 + O(\log \log N / \log N)$. Neither report cites the other; the two were produced by the same pipeline on the same day. The following ascent-sequence reports of the same collection treat other models and share no theorem with this report:

- `a202058-ascent-000-growth`: ascent sequences avoiding 000;
- `a294220-ascent-multiplicity-caps`: ascent sequences with capped multiplicities;
- `a202061-ascent-120-deficit` and `a202062-ascent-201-enumeration`: other avoidance classes;
- the batch-102 sibling `a336070-weak-ascents` (weak and difference ascent sequences, not their self-modified tables).

No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status.

Notation. The manuscript reuses many letters. The table fixes each meaning; no symbol was renamed.

Symbol	Meanings (where)	Not to be read as
N, n	table size (entry sum) N ; n the OEIS row index of A098568, $N = n + 1$ (1.10), and integer sizes in Section 9; n_0 the seed size (9.12); the letter N of “ N -free” is the poset shape	the OEIS index of A098569 ($b_N = A098569(N-1)$)
T, t	$T(N, m)$ the count (1.3); T also the bound $ t \leq T$ on the real dimension mark t (Theorems 2.1, 5.1)	the mark bound as a count
H	$H_N(x)$ the comparison phase (3.1) (Sections 3, 6, 8); $H(x)$ the leading binary penalty (7.5); $H_0(x) = -2N(N-x)/x^2$ (Section 8)	H or H_0 as H_N (all three occur in Section 8)
B	$B(z, h)$ the series (1.6); $B_N(t)$ the marked count (2.1); $B = N - 1$ in (7.4); $B_0 = -F/2 + 1/u$ (8.5); $\mathcal{B}(s, q)$ a gamma ratio (proof of Lemma 3.1)	B_0 as B or as B_N at $N = 0$ (the source flags this one)
C, c	$C(z, h)$ (1.7); c_N binary counts; C_ℓ the contractions (2.8), $C_\ell(y)$ at real size; $C_K, C, c, c', c_1, c_2, c_3$ unspecified constants; C_{bad} a cell count (8.14)	C_K (Theorem 9.1) as a contraction
D, d	D_N the random dimension (Section 5); $D = u + 4 + 2/u$ (8.5); D_{bad} (8.13); D_K an exponent (4.2) and an error constant (proof of Theorem 9.1); d the difference parameter of [5], the degree in (7.10), and $d = -B + Bs + qv$ (Section 7)	
E	$\mathbb{E}$ expectation; $E = 2u + 6 - 4/u^2$ (8.5)	
F	$F_K(y) = \log S_K(y)$ (9.2); F_0, F_- (proof of Theorem 9.2); $F = (u+1)(u+2)$ (8.5)	
G	G_k the binary numerator coefficients (7.10); $G = (u+2)(3u+4)$ (8.5)	
K, k	K the fixed expansion order; K_n the random column index of A098568 (end of Section 5); k, k_j indices	one convention: (2.9) sums $\ell < K$ with error N^{-K} , (7.12) sums $k \leq K$ with error $L^{4K+4}N^{-K-1}$

Symbol	Meanings (where)	Not to be read as
P	$P(u) = u^2 + 4u + 2$ (6.3); $P_y(x) = M(x) + y - x$ (Section 9); polynomials $P_j(s)$ (9.21); $\mathbb{P}$ probability	
R	$R(s, q)$ (3.3); $R_K(z)$ (4.1); $\mathcal{R}(u)$ (8.1); the window $ z \leq R$ (Theorem 5.1); $\mathbb{R}$ the reals	
h, g	h the dimension mark of $B(z, h)$, $C(z, h)$; h_N the binary penalty (7.2); the mesh $h = \sqrt{A}$ (Lemma 4.1); g_N (7.1), $g_{\pm}$ (proof of Theorem 9.2)	
u, v, w	$u = u_N$ of (6.2), $u(y)$ at real size; $v = 2w$ with $w = W_0(\sqrt{N/2})$ (Section 8.1); v of (9.11) and $w = \log \Lambda$ (Section 9.3); $w \geq 1$ in (3.14); the weight w (Section 4); integration variables v	the w, v of Section 8.1 as those of Section 9
z	the formal variable of $B(z, h)$; the standardized $z = \sqrt{A}(x - r)$ (Sections 4, 5, 7); the seeds z_b, z_c (9.12)	
s, q	$s = M(x)$ (Section 3); the binary saddle s (7.17); an integration variable (7.4); $q = N - x$ the residual; Q_k (7.11)	
X, Λ	the threshold X (9.1), $\Lambda = \log X$; a residual cell X (8.12)	
ε	$\varepsilon = N^{-1/2}$ (Section 4); the window exponent $\varepsilon < 1/6$ (proof of Theorem 7.1)	

What is not claimed (collected from the source): the exact enumeration, the word/table bridge, the rigid binary subclass and the multiplicity substitution are prior; the $N \log N$ scale is prior, the finer logarithmic expressions are Kotešovec's OEIS postings, and a logarithmic-scale equivalence does not give lower-order terms with additive error $o(N)$; historical N -freeness is in the directed covering graph and must not be conflated with induced-subposet N -avoidance or the series-parallel class, so results are stated in table and word language; the 1989 asymptotic theorems concern the larger class of covering-graph N -free posets; results on Fishburn matrices [9] do not transfer automatically; only $d = 0$ and $d = 1$ are treated; the expansion order is fixed and marks are bounded and real, with no convergence as $K \rightarrow \infty$, no uniformity in a growing order and no claim about stationary points outside I_N ; O -constants and thresholds are non-effective, and remainders are not differentiable in t or N ; there is no unconditional rule $\nu_b(X) = \lceil y_K(X) \rceil$, and floating point certifies no integer threshold; the limits (2.12) cannot replace the exact C_1, C_2 ; the collision expectations of Section 8 are an interpretation, not a proof; the numerical diagnostics are finite consistency checks, reproducible only for the recorded toolchain, the manifest proves integrity, not authorship or correctness, and the guard tests are not a security proof; the source search was bounded, with no worldwide novelty or priority, no external review and no formal verification, and its sources were read as web-extracted text, not page images. The delivered `SOURCES.md` also records that a 2004 bibliography gives the 1989 paper's pages as 219–232; the correct range is 219–225. Open questions are collected in Section 11.1.

2 The exact gamma saddle and the all orders theorem

For a real dimension mark t , let

$$B_N(t) = \sum_{m=1}^N T(N, m) e^{tm}, \quad B_N(0) = b_N. \quad (2.1)$$

The real phase on $1 \leq x \leq N$ is

$$f_N(x) = \log \Gamma(M(x) + N - x) - \log \Gamma(N - x + 1) - \log \Gamma(M(x)). \quad (2.2)$$

It agrees *exactly* with the summands: $e^{f_N(m)} = T(N, m)$. Replacing this phase by only its leading logarithmic approximation would lose an unbounded exponential factor; Section 6 quantifies this issue.

Theorem 2.1 (Exact saddle expansion). *Fix a finite $T \geq 0$ and an integer $K \geq 1$. Write $L = \log N$ and*

$$I_N = [N/(2L), 4N/L].$$

For all sufficiently large N , uniformly for real $|t| \leq T$, there is exactly one root $r = r_N(t)$ in I_N of

$$f'_N(r) + t = 0. \quad (2.3)$$

It is the unique global maximizer of $f_N(x) + tx$ on $[1, N]$. Put

$$A = -f''_N(r), \quad \sigma = A^{-1/2}, \quad \lambda_j = f_N^{(j)}(r)A^{-j/2} \quad (j \geq 3). \quad (2.4)$$

Then

$$r = \frac{2N}{L - 2 \log L + \log 2 + 2 - t + O_T(\log L/L)}, \quad (2.5)$$

$$A = \frac{2N}{r^2} (1 + O_T(L^{-1})) \sim \frac{L^2}{2N}, \quad (2.6)$$

$$\lambda_j = O_{j,T}(N^{1-j/2}). \quad (2.7)$$

Define $C_0 = 1$ and, for $\ell \geq 1$,

$$C_\ell = \sum_{\substack{k_3, \dots, k_{2\ell+2} \geq 0 \\ \sum_{j=3}^{2\ell+2} (j-2)k_j = 2\ell}} \left(\sum_{j=3}^{2\ell+2} jk_j - 1 \right)!! \prod_{j=3}^{2\ell+2} \frac{(\lambda_j/j!)^{k_j}}{k_j!}. \quad (2.8)$$

Here $(-1)!! = 1$. This is a finite sum, its double factorial is an even Gaussian moment, and $C_\ell = O_{\ell,T}(N^{-\ell})$. The marked count satisfies

$$B_N(t) = e^{f_N(r) + tr} \sqrt{\frac{2\pi}{A}} \left(\sum_{\ell=0}^{K-1} C_\ell + O_{K,T}(N^{-K}) \right). \quad (2.9)$$

The theorem permits each *fixed* K and bounded real t . It asserts neither convergence as $K \rightarrow \infty$ nor an error uniform in an order growing with N . It also does not assert the absence of smaller stationary points outside I_N .

For $f_j = f_N^{(j)}(r)$, the first corrections are

$$C_1 = \frac{f_4}{8A^2} + \frac{5f_3^2}{24A^3}, \quad (2.10)$$

$$C_2 = \frac{f_6}{48A^3} + \frac{7f_3f_5}{48A^4} + \frac{35f_4^2}{384A^4} + \frac{35f_3^2f_4}{64A^5} + \frac{385f_3^4}{1152A^6}. \quad (2.11)$$

In particular,

$$C_1 \sim \frac{1}{24N}, \quad C_2 \sim \frac{1}{1152N^2}. \quad (2.12)$$

The limits in (2.12) cannot replace the exact coefficients in (2.9) at its stated power accuracy. Their logarithmic relative corrections are larger than the next inverse- N power. For example, in the bounded computations through $N = 1000$, the exact C_1 at $t = 0$ is still negative.

Remark 2.2 ([write] How slowly NC_1 approaches $1/24$). Added 5 October 2026, batch 102. The intake evaluated C_1 at $t = 0$ from (2.10), with the exact gamma phase (2.2) and its derivatives at the exact saddle, in 40- to 60-digit floating point (not interval arithmetic). It found

N	10^3	10^4	10^5	10^6	10^9	10^{12}	10^{20}
NC_1	-0.0113	-0.0016	0.0056	0.0109	0.0206	0.0257	0.0321

against the limit $1/24 = 0.0417$. On a logarithmic grid of 51 sizes from 10 to 10^6 , NC_1 is negative up to $N = 15,849$ and positive from $N = 19,953$ on, with its least value -0.0217 near $N = 50$; bisection places the sign change between $N = 15,864$ ($NC_1 \approx -1.1 \times 10^{-7}$) and $N = 15,865$ ($NC_1 \approx +1.1 \times 10^{-7}$). So $C_1 < 0$ until N lies between 10^4 and 10^6 , as the source’s caution that “a short finite range need not yet show the limiting sign” (Section 10) anticipates. The reason is visible in the proof: by (3.11), each of the two terms of (2.10) carries a relative error $O(1/L)$, so $NC_1 = -\frac{3}{8} + \frac{5}{12} + O(1/L) = \frac{1}{24} + O(1/L)$, the difference of two terms of size about 0.4; at $N = 10^6$, $1/L \approx 0.07$. The table is a finite uncertified evaluation, not a theorem (Question 1).

The rest of the proof is split into global localization and a lattice expansion. This makes explicit why neither an isolated formal saddle calculation nor an unproved replacement of the sum by an integral is enough.

3 Global localization and differentiable estimates

The analytic inputs are the standard positive-real Stirling, digamma and polygamma estimates [11]. They also follow by integral comparison from

$$\psi_j(z) = (-1)^{j+1} j! \sum_{k \geq 0} (z+k)^{-j-1} \quad (j \geq 1, z > 0),$$

with the digamma asymptotic $\psi(z) = \log z + O(z^{-1})$. We will use fixed derivative orders only.

3.1 A comparison valid up to the endpoint

Define

$$H_N(x) = (N-x) \left\{ \log \frac{x^2}{2(N-x)} + 1 \right\}, \quad H_N(N) = 0. \quad (3.1)$$

For every fixed $a > 0$,

$$f_N(x) = H_N(x) + O_a(L^2), \quad aN/L \leq x \leq N. \quad (3.2)$$

To prove this uniformly, including $x = N$, set $s = M(x)$ and $q = N - x$. The exact integral representation

$$\log \Gamma(s+q) - \log \Gamma(s) = q \log s + R(s, q), \quad (3.3)$$

$$R(s, q) = \int_0^q (\psi(s+v) - \log s) dv = O\left(\frac{q(q+1)}{s}\right) \quad (3.4)$$

holds in this range. Also

$$q \log s = q \log(x^2/2) + O(N/x), \quad \log \Gamma(q+1) = q \log q - q + O(\log(q+2)),$$

where $q \log q = 0$ at zero. The latter bound follows from Stirling for $q \geq 1$ and continuity on $[0, 1]$. These estimates give (3.2), without any assumption that q is large at the endpoint.

For each fixed $\alpha > 0$,

$$H_N(\alpha N/L) = N[L - 2 \log L + 2 \log \alpha - \log 2 + 1 - \alpha] + O_\alpha(N \log L/L). \quad (3.5)$$

In particular, at $x_* = 2N/L$,

$$f_N(x_*) + tx_* = N[L - 2 \log L + \log 2 - 1] + O_T(N \log L/L).$$

Lemma 3.1 (Global exclusion). *Uniformly for $|t| \leq T$, for all sufficiently large N ,*

$$\sup_{x \in [1, N] \setminus I_N} \{f_N(x) + tx\} \leq f_N(x_*) + tx_* - cN, \quad (3.6)$$

where, for example, $c = 1/4$ is admissible eventually.

Proof. For the lower tail, let

$$\mathcal{B}(s, q) = \frac{\Gamma(s+q)}{\Gamma(s)\Gamma(q+1)}.$$

For $s \geq 1, q \geq 0$, this function is nondecreasing in both variables because

$$\partial_s \log \mathcal{B} = \psi(s+q) - \psi(s) \geq 0, \quad \partial_q \log \mathcal{B} = \psi(s+q) - \psi(q+1) \geq 0.$$

Here $s+q \geq q+1$ and ψ is increasing. If $x \leq aN/L$, $a = 1/2$, then

$$f_N(x) \leq \log \mathcal{B}(M(aN/L), N) = N[L - 2 \log L + 2 \log a - \log 2 + 1] + O(L^2).$$

The mark is $O_T(N/L)$ in this tail. Comparison with x_* gives a gap $(2 \log 4 - 2)N + o(N)$.

For the upper tail, direct differentiation gives

$$H'_N(x) = 2N/x - 2 - 2 \log x + \log(2(N-x)), \quad (3.7)$$

$$H''_N(x) = -2N/x^2 - 2/x - 1/(N-x) < 0. \quad (3.8)$$

At $x = 4N/L$, $H'_N(x) + t = -L/2 + 2 \log L + O_T(1) < 0$. Thus $H_N(x) + tx$ is decreasing thereafter. Equations (3.2) and (3.5) give an upper-tail gap $(2 - 2 \log 2)N + o(N)$. Both constants exceed $1/4$, proving the assertion. $\square$

3.2 Derivatives from an explicit remainder

We do not differentiate the unstructured error in (3.2). On a fixed proportional interval

$$aN/L \leq x \leq bN/L, \quad 0 < a < b < \infty,$$

we have $q \asymp N$, $s \asymp x^2$ and $q/s = o(1)$. The integral (3.3), differentiated at fixed orders, gives

$$\begin{aligned} \partial_s^a R &= O_a(q(q+1)s^{-a-1}), \\ \partial_s^a \partial_q R &= O_a((q+1)s^{-a-1}), \\ \partial_s^a \partial_q^b R &= O_{a,b}(s^{-a-b+1}) \quad (b \geq 2). \end{aligned}$$

For example,

$$\frac{d^a}{ds^a}(\psi(s+v) - \log s) = O_a((v+1)s^{-a-1})$$

is uniform for $0 \leq v \leq q$. Composition with the quadratic $s(x)$ and linear $q(x)$ shows

$$\frac{d^j}{dx^j} R(s(x), q(x)) = O_j(N^2 x^{-j-2}).$$

Terms with zero q derivatives have this size, those with one have size at most Nx^{-j-1} , and terms with $b \geq 2$ have size at most x^{-j-b+2} ; all are bounded by the displayed quantity in this range. The term $q \log(1 + 1/x)$ contributes $O_j(Nx^{-j-1})$, while the differentiated Stirling remainder for q contributes $O_j(N^{-j})$. Consequently,

$$f_N^{(j)}(x) - H_N^{(j)}(x) = O_j\left(\frac{N}{x^{j+1}} + \frac{N^2}{x^{j+2}} + N^{-j}\right), \quad j \geq 1. \quad (3.9)$$

All orders are fixed and the constants may depend on a, b, j .

For $j \geq 2$, the explicit derivative is

$$H_N^{(j)}(x) = \frac{2(-1)^{j-1}(j-1)!N}{x^j} + \frac{2(-1)^{j-1}(j-2)!}{x^{j-1}} - \frac{(j-2)!}{(N-x)^{j-1}}. \quad (3.10)$$

It follows that

$$f_N^{(j)}(x) = \frac{2(-1)^{j-1}(j-1)!N}{x^j} (1 + O_j(L^{-1})), \quad j \geq 2. \quad (3.11)$$

No derivative estimate for $q \asymp N$ has been used at $x = N$.

3.3 The unique dominant saddle and lattice tails

On I_N , (3.9) implies

$$f'_N(x) = 2N/x - 2 - 2\log x + \log(2(N-x)) + O(L^3/N).$$

At its left and right endpoints, respectively, $f'_N(x) + t$ equals $3L + 2\log L + O_T(1) > 0$ and $-L/2 + 2\log L + O_T(1) < 0$. Equation (3.11) gives strict concavity throughout I_N . This proves existence and uniqueness there; Lemma 3.1 proves global dominance.

Set $y = 2N/r$. The saddle equation becomes

$$y = L - 2\log y + \log 2 + 2 - t - \log(1 - 2/y) + O(L^3/N).$$

First $y \sim L$, and substitution then gives (2.5). The other estimates in Theorem 2.1 follow from (3.11). In addition,

$$-c_2 L^2/N \leq f''_N(x) \leq -c_1 L^2/N \quad (x \in I_N), \quad (3.12)$$

so with $\Phi(x) = f_N(x) + tx$,

$$\Phi(x) \leq \Phi(r) - c_3 A(x-r)^2 \quad (x \in I_N). \quad (3.13)$$

The nearest integer to r has phase $\Phi(r) + O(A)$; more strongly, the $\asymp \sigma$ integers within a fixed multiple of σ give a lower bound $c\sigma e^{\Phi(r)}$ for the sum. The global tail is at most $Ne^{\Phi(r)-cN}$. Gaussian lattice-tail comparison therefore gives, for $w \geq 1$,

$$\frac{\sum_{|m-r|>w\sigma} e^{\Phi(m)}}{B_N(t)} \leq Ce^{-c'w^2} + CN e^{-c'N}, \quad (3.14)$$

when the sum is over $1 \leq m \leq N$, with constants adjusted if the window crosses the ends of I_N . An analogous bound with a fixed polynomial in $|m - r|/\sigma$ follows by integrating a Gaussian polynomial tail. These estimates will justify moments, lattice limits and the rarer binary event.

4 The discrete expansion and its error

Write $z = \sqrt{A}(x - r)$ and $\varepsilon = N^{-1/2}$. On the window $|x - r| \leq \sqrt{N}$, whose standardized width is $\asymp L$, Taylor's theorem through order $2K + 1$ yields

$$\Phi(x) - \Phi(r) = -\frac{z^2}{2} + \sum_{j=3}^{2K+1} \frac{\lambda_j z^j}{j!} + R_K(z), \quad |R_K(z)| \leq C_K N^{-K} |z|^{2K+2}. \quad (4.1)$$

The derivative of order $2K + 2$ is controlled on a fixed enlargement of this window, which eventually remains inside I_N . The entire nonquadratic exponent is $O(L^3/\sqrt{N}) = o(1)$ here.

Let $a_j = N^{(j-2)/2} \lambda_j$. The a_j are uniformly bounded. Expand

$$\exp \left(\sum_{j=3}^{2K+1} \varepsilon^{j-2} a_j z^j / j! \right)$$

as a polynomial in ε through degree $2K - 1$. The coefficient of ε^w is a finite sum of monomials with $\sum (j - 2)k_j = w$; its parity in z is the parity of w . Finite Taylor's theorem in ε , together with (4.1), bounds the omitted part after multiplication by $e^{-z^2/2}$ by

$$C_K N^{-K} (1 + |z|)^{D_K} e^{-z^2/2} \quad (4.2)$$

for some fixed D_K . The uniform $o(1)$ exponent correction permits a fixed multiplicative bound. The Gaussian-polynomial lattice sum of (4.2) is $O_K(N^{-K}/\sqrt{A})$, uniformly in the fractional part of r .

Lemma 4.1 (Gaussian lattice transfer). *For each fixed polynomial p and $h \rightarrow 0^+$, uniformly in real r ,*

$$h \sum_{m \in \mathbb{Z}} p(h(m - r)) e^{-h^2(m-r)^2/2} = \int_{\mathbb{R}} p(z) e^{-z^2/2} dz + O_p(e^{-c/h^2}). \quad (4.3)$$

Proof. The Fourier transform of a polynomial times $e^{-z^2/2}$ is another polynomial times a Gaussian. Poisson summation evaluates the left side as the zero-frequency integral plus terms at frequencies $2\pi k/h$, multiplied by unit-modulus factors depending on r . Their absolute sum is bounded by a polynomial in h^{-1} times $e^{-2\pi^2/h^2}$. Absorbing that polynomial into a smaller exponential constant proves (4.3). This argument also proves uniformity in the lattice shift. $\square$

Although the expansion coefficients depend on N, t , each fixed order contains only finitely many monomials with bounded coefficients, so the lemma applies uniformly term by term. In the present problem $h = \sqrt{A}$ and $h^{-2} \asymp N/L^2$. Thus the error in (4.3) is smaller than every fixed power of N^{-1} . Extending the local polynomial sums to all integers costs $O(e^{-cL^2})$ relative to the main scale, by (3.14) and Gaussian tails. The same is true of the discarded actual summands. These errors, too, are smaller than every fixed power of N^{-1} .

Odd-weight terms now have zero integral and only an exponentially small lattice discrepancy. At weight 2ℓ , the normalized Gaussian integral of $z^{\sum jk_j}$ is $(\sum jk_j - 1)!!$. This is exactly (2.8), and proves (2.9) with its stated remainder.

The leading derivative limits are

$$\lambda_j \sim 2^{1-j/2}(-1)^{j-1}(j-1)!N^{1-j/2}. \quad (4.4)$$

Using $j = 3, 4$ in (2.10) gives $NC_1 \rightarrow -3/8 + 5/12 = 1/24$; the corresponding substitution in (2.11) gives $N^2C_2 \rightarrow 1/1152$. This proves (2.12). More generally, the same Gaussian-contraction rule provides every coefficient without fitting numerical data. One may recognize their leading constants as the Stirling-series coefficients for a gamma integral of shape $2N$, but this recognition is not needed in the proof or in the algorithm.

Remark 4.2 (What the error statement means). The constants in all $O_{K,T}$ terms are existence constants for fixed K, T . The proof does not provide usable numerical thresholds at which a prescribed finite error is guaranteed. It also does not justify differentiation of the asymptotic remainder in t or N . Whenever derivatives are needed below, we estimate the exact phase and finite coefficient formulas directly.

5 Dimension and the distribution of the rows of A098568

Choose a size- N table with probability proportional to e^{tm} , and let D_N denote its dimension. Thus

$$\mathbb{P}_t(D_N = m) = \frac{T(N, m)e^{tm}}{B_N(t)}.$$

In this section $r, A, \sigma, \lambda_j, C_j$ are evaluated at the same t , and all assertions are uniform on each fixed interval $|t| \leq T$.

Theorem 5.1 (Local expansion and Gaussian limit). *Let $\varphi(z) = (2\pi)^{-1/2}e^{-z^2/2}$ and $z = \sqrt{A}(m-r)$. For every fixed $R < \infty$, uniformly over support points with $|z| \leq R$,*

$$\mathbb{P}_t(D_N = m) = \sqrt{A}\varphi(z) \left[1 + \frac{\lambda_3 z^3}{6} + \frac{\lambda_4 z^4}{24} + \frac{\lambda_5^2 z^6}{72} - C_1 + O_R(N^{-3/2}) \right]. \quad (5.1)$$

Moreover, with probabilities set to zero off the support,

$$\sup_{m \in \mathbb{Z}} \left| \sigma \mathbb{P}_t(D_N = m) - \varphi\left(\frac{m-r}{\sigma}\right) \right| \rightarrow 0, \quad (5.2)$$

and

$$\frac{D_N - r}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1). \quad (5.3)$$

The support $\{1, \dots, N\}$ has span one.

Proof. Taylor expansion of the exact numerator at fixed z , followed by division by (2.9) through the first correction, gives (5.1). The term $-C_1$ is its normalizing correction. The expansion is centered at the saddle, rather than the exact mean, which explains the cubic z^3 term.

For $m \in I_N$, (3.13) and the denominator lower bound imply $\sigma \mathbb{P}_t(D_N = m) \leq Ce^{-cz^2}$. Outside I_N , the corresponding bound is Ce^{-cN} . The Gaussian itself tends uniformly to zero at those remote support points and outside $\{1, \dots, N\}$. First apply the compact- z formula and then let $R \rightarrow \infty$ to prove (5.2). For the distributional limit, the same tail bound gives tightness, and the compact sums become Gaussian Riemann sums with mesh $1/\sigma \rightarrow 0$. This proves (5.3); an unweighted supremum local error alone would not suffice. Finally, every $T(N, m)$ is positive, so there is no sublattice obstruction. $\square$

Proposition 5.2 (Mean and variance corrections). *At the same saddle,*

$$\mathbb{E}_t D_N = r + \frac{f_3}{2A^2} + O(\sigma N^{-3/2}), \quad (5.4)$$

$$\text{Var}_t D_N = A^{-1} + \frac{f_4}{2A^3} + \frac{f_3^2}{A^4} + O(A^{-1}N^{-2}). \quad (5.5)$$

In particular, at $t = 0$,

$$\mathbb{E} D_N \sim \frac{2N}{\log N}, \quad \text{Var } D_N \sim \frac{2N}{(\log N)^2}.$$

The error in (5.4) is $O(1/(N \log N))$.

Proof. Repeat the proof of Section 4 with factors z and z^2 . The polynomial tail bounds give uniform integrability, so these weighted expansions are legitimate. Parity cancels the next inappropriate weights and yields

$$\mathbb{E}_t z = \lambda_3/2 + O(N^{-3/2}), \quad \mathbb{E}_t z^2 = 1 + \lambda_4/2 + 5\lambda_3^2/4 + O(N^{-2}).$$

For example the unnormalized z^2 correction is $15\lambda_4/24 + 105\lambda_3^2/72$; subtracting C_1 gives the displayed raw moment. Subtracting $(\mathbb{E}_t z)^2$ gives $\text{Var}_t z = 1 + \lambda_4/2 + \lambda_3^2 + O(N^{-2})$. Rescaling proves both formulas. This proof does not differentiate a real-mark remainder. $\square$

Centering and scaling by the exact mean and standard deviation gives the same normal limit, since their standardized corrections tend to zero. All higher fixed moments and local corrections are obtainable from the weighted-contraction rule and finite series division.

At moderate distances the relative Gaussian approximation holds uniformly whenever $|z| = o(N^{1/6})$. Indeed, these displacements are $o(r)$ and the first nonquadratic exponent is $O(N^{-1/2}|z|^3) = o(1)$ on the intervening proportional interval. Beyond this range it is safer to retain the exact shape: for any support point,

$$\mathbb{P}_t(D_N = m) = \sqrt{\frac{A}{2\pi}} e^{f_N(m) + tm - f_N(r) - tr} \left(\sum_{\ell=0}^{K-1} C_\ell + O_{K,T}(N^{-K}) \right)^{-1}. \quad (5.6)$$

This is the exact numerator divided by the proved denominator; it does not extend a Gaussian relative approximation into a large-deviation region.

[write] **Added 5 October 2026, batch 102.** The moderate-distance claim of the preceding paragraph is argued in two sentences. The following proposition states it with an explicit error and proves it from (3.11), (2.6) and (5.6). The claim “all higher fixed moments and local corrections are obtainable” remains a further question (Question 6).

Proposition 5.3 ([write] Moderate deviations of the dimension). *Fix $T < \infty$. Uniformly for real $|t| \leq T$ and for integers m with $|m - r| \leq r/2$, with $z = \sqrt{A}(m - r)$,*

$$\mathbb{P}_t(D_N = m) = \sqrt{A} \varphi(z) \exp\left(O_T(|z|^3 N^{-1/2} + N^{-1})\right).$$

In particular $\mathbb{P}_t(D_N = m) = \sqrt{A} \varphi(z)(1 + o(1))$ uniformly over $|z| \leq \omega_N N^{1/6}$, for every sequence $\omega_N \rightarrow 0$.

Proof. By (2.5), $r = (2N/L)(1+o(1))$ uniformly in $|t| \leq T$, so for all large N every x with $|x-r| \leq r/2$ lies in $I_N = [N/(2L), 4N/L]$; in particular such an integer m is a support point. On I_N , (3.11) with $j = 3$ (constants depending only on the fixed interval) gives $|f_N'''(x)| \leq C'N/x^3 \leq 8C'L^3/N^2$, and (2.6) gives $A \geq L^2/(4N)$, hence $A^{-3/2} \leq 8N^{3/2}/L^3$, for all large N , uniformly in t . Let $\Phi(x) = f_N(x) + tx$. Since $\Phi'(r) = 0$, $\Phi''(r) = -A$ and $\Phi''' = f_N'''$, Taylor's theorem with the Lagrange remainder gives, for some ξ between r and m ,

$$\Phi(m) - \Phi(r) = -\frac{z^2}{2} + \rho, \quad |\rho| = \frac{|f_N'''(\xi)| |m-r|^3}{6} \leq \frac{8C'L^3}{6N^2} \cdot \frac{8N^{3/2}}{L^3} |z|^3 = \frac{32C'}{3} |z|^3 N^{-1/2}.$$

Formula (5.6) with $K = 1$ (so the bracket is $C_0 + O_T(N^{-1}) = 1 + O_T(N^{-1})$) gives $\mathbb{P}_t(D_N = m) = \sqrt{A/(2\pi)} e^{-z^2/2+\rho}(1+O_T(N^{-1}))$, which is the first assertion. For the second, $\sigma = A^{-1/2} \leq 2N^{1/2}/L$, so $|z| \leq \omega_N N^{1/6}$ implies $|m-r| \leq 2\omega_N N^{2/3}/L \leq r/2$ for large N , while $|z|^3 N^{-1/2} \leq \omega_N^3 \rightarrow 0$. $\square$

Thus $|z| = o(N^{1/6})$ is the range in which the cubic term $\lambda_3 z^3/6$ of (5.1), of size $\asymp |z|^3 N^{-1/2}$, stays small; beyond it the exact shape (5.6) should be used, as the source says.

To apply the theorem to A098568, normalize its row n by its row sum. If the random column index is K_n , then $N = n + 1$ and $D_N = K_n + 1$. Thus its saddle center is $r_{n+1}(0) - 1$, its leading variance is $2(n+1)/(\log(n+1))^2$, and (5.2) is a full-span local Gaussian answer to the row-distribution question. The column shift affects the center by one and is not to be dropped in a precise local statement.

[write] **Added 5 October 2026, batch 102: the question of A098568.** OEIS A098568 asks, in a comment read on 5 October 2026, “How do the terms of row k tend to be distributed as k grows?” (its row index is our n). Theorem 5.1, Proposition 5.2 and Proposition 5.3 answer it in the sense above: after normalization by the row sum b_{n+1} , the row is a full-span local Gaussian law in the column index $K_n = D_{n+1} - 1$, with the corrections (5.1), relative accuracy up to $|z| = o(n^{1/6})$ and the exact shape (5.6) beyond. No repository report asked this question; it is answered here for the first time in ProveIt, and the source claims no priority beyond its bounded search. The leading equivalents are slow: at $n = 999$ ($N = 1000$) the row has mean column index 314.606 and variance 36.557 (Section 10), against $2N/\log N - 1 = 288.5$ and $2N/(\log N)^2 = 41.9$, while the saddle centre $r - 1 = 314.514$ and the corrected moments of Proposition 5.2 agree with the exact values to about 7×10^{-6} and 1.5×10^{-6} (intake spot checks, uncertified).

6 An elementary relative carrier with the log square factor

At integer $q = N - m$ and $M = M(m)$,

$$\log \frac{T(N, m)}{M^q/q!} = \sum_{j=0}^{q-1} \log(1 + j/M) = \frac{q(q-1)}{2M} + O(q^3/M^2). \quad (6.1)$$

When $m \sim 2N/\log N$, the main correction is asymptotic to $(\log N)^2/4$, whereas $q^3/M^2 = O((\log N)^4/N) = o(1)$. The naive Poissonized cell count therefore misses a factor that grows without bound. An $O(\log^2 N)$ error in an exponent cannot prove a relative equivalent.

There is nevertheless a simple implicit parametrization that retains the necessary factor. For $N > 0$, let $u = u_N$ be the unique positive solution of

$$u(u+2)e^u = 2N, \quad (6.2)$$

and put

$$P(u) = u^2 + 4u + 2. \quad (6.3)$$

The left side of (6.2) is strictly increasing from zero to infinity; also $u \sim \log N$.

Theorem 6.1 (Elementary count carrier). *As $N \rightarrow \infty$,*

$$b_N = \sqrt{\frac{2}{P(u)}} \exp\left(\frac{Nu(u+1)}{u+2} + \frac{u}{2} + \frac{u^2}{4}\right) \left(1 + O\left(\frac{(\log N)^4}{N}\right)\right). \quad (6.4)$$

The implicit u is retained exactly in this formula.

Proof. Set $x_0 = 2N/(u+2) = ue^u$ and $q_0 = N - x_0 = ux_0/2$. Equation (6.2) gives $\log(x_0^2/(2q_0)) = u$, so $H'_N(x_0) = 0$. Direct differentiation shows

$$A_0 = -H''_N(x_0) = \frac{u+4+2/u}{x_0}, \quad A_0 q_0 = P(u)/2. \quad (6.5)$$

The differentiated estimates in Section 3 give

$$f'_N(x_0) = O(u^3/N), \quad f''_N(x_0) = -A_0(1 + O(u^2/N)).$$

The saddle displacement is therefore $r - x_0 = O(u)$, and Taylor's theorem with the uniform curvature bounds gives

$$f_N(r) - f_N(x_0) = O(u^4/N), \quad A/A_0 = 1 + O(u^2/N). \quad (6.6)$$

For the value, use the uniform gamma quotient and Stirling expansions

$$\begin{aligned} \log \Gamma(M+q) - \log \Gamma(M) &= q \log M + \frac{q(q-1)}{2M} + O(q^3/M^2), \\ \log \Gamma(q+1) &= (q+1/2) \log q - q + \frac{1}{2} \log(2\pi) + O(q^{-1}). \end{aligned}$$

At $x = x_0$, write $M = x_0^2(1+1/x_0)/2$. Then $q_0 \log(1+1/x_0) = u/2 + O(u^2/N)$ and $q_0(q_0-1)/(2M) = u^2/4 + O(u^3/N)$, so

$$f_N(x_0) = q_0(u+1) + u/2 + u^2/4 - \frac{1}{2} \log(2\pi q_0) + O(u^4/N).$$

Apply Theorem 2.1 with $K = 1$, use (6.6), and combine the Gaussian prefactor using (6.5). Since $q_0(u+1) = Nu(u+1)/(u+2)$, this proves (6.4). $\square$

In particular, the proof gives the coarse consequence

$$\log b_N = N[\log N - 2 \log \log N + \log 2 - 1 + o(1)]. \quad (6.7)$$

As explained in Section 1, the leading scale is prior and finer logarithmic expressions are already recorded in OEIS. The relative statement (6.4) is stronger than merely exponentiating a logarithmic-scale equivalence. Replacing u by a fixed finite logarithmic expansion generally destroys its relative accuracy because the remaining error is multiplied by the large size in the exponent.

7 Binary tables as a rare event

For the uniform size- N model, define

$$p_N = \mathbb{P}(\text{the table is binary}) = c_N/b_N.$$

We work at the unmarked denominator saddle $r = r_N(0)$ throughout this section. No convergence-in-distribution result will be used to estimate this rare probability.

7.1 The exact penalty and its derivatives

On the dominant interval, set

$$g_N(x) = \log \Gamma(M(x) - x + 1) - \log \Gamma(N - x + 1) - \log \Gamma(M(x) - N + 1), \quad (7.1)$$

$$h_N(x) = g_N(x) - f_N(x). \quad (7.2)$$

All gamma arguments here are positive for sufficiently large N , because $M(x) \asymp N^2/L^2 \gg N$ on I_N . At feasible integer dimensions,

$$e^{h_N(m)} = \frac{U(N, m)}{T(N, m)} = \prod_{j=0}^{q-1} \frac{M - m - j}{M + j}. \quad (7.3)$$

Infeasible dimensions have zero binary weight and require no smooth continuation.

There is a useful identity valid for real x near the saddle. With $B = N - 1$ and $q = N - x$,

$$h_N(x) = -Bq \int_0^1 \int_0^1 \psi_1(M - B + Bs + qv) ds dv. \quad (7.4)$$

This is the second finite difference of $\log \Gamma$ over a rectangle; its corner $M - B + q = M - x + 1$ verifies the four gamma terms.

Define

$$H(x) = -\frac{(N-x)(N-1)}{M(x)}, \quad J(x) = -\frac{x(N-x)(N-1)}{2M(x)^2}. \quad (7.5)$$

Uniformly on fixed proportional neighborhoods of r ,

$$h_N^{(j)}(x) = H^{(j)}(x) + J^{(j)}(x) + O_j\left(\frac{L^6}{N^2 x^j}\right), \quad j \geq 0. \quad (7.6)$$

Here and below each derivative order is fixed. We supply the uniform analytic justification. In a small disk $|z - x| \leq cx$ about such a real x , $M(z)$ stays in a fixed right-half-plane sector and $|M(z)| \asymp x^2$; each argument in (7.4) differs from it by $O(N) = o(x^2)$. If $d = -B + Bs + qv$, the sectorial trigamma expansion gives

$$\psi_1(M + d) = \frac{1}{M} + \frac{1/2 - d}{M^2} + O(N^2/|M|^3)$$

uniformly in that disk and in $s, v \in [0, 1]$. The average of d is $(q - B)/2 = (1 - x)/2$, so the average of $1/2 - d$ is $x/2$. Multiplication by $-Bq$ gives exactly $H + J$ and an analytic remainder $O(N^4/x^6) = O(L^6/N^2)$. Cauchy's estimate on a slightly smaller disk gives the additional $O_j(x^{-j})$ for derivatives. Thus (7.6) is not a formal differentiation of a real-axis error bound.

It follows in particular that

$$h_N^{(j)}(x) = O_j(L^{j+2}N^{-j}), \quad (7.7)$$

$$J^{(j)}(x) = O_j(L^{j+3}N^{-j-1}). \quad (7.8)$$

Let

$$\eta_j = h_N^{(j)}(r)A^{-j/2} \quad (j \geq 1). \quad (7.9)$$

Then $\eta_j = O_j(L^2N^{-j/2})$ and $\eta_1 = o(1)$. Although $h_N(r) = -O(L^2)$, the penalty varies by $o(1)$ across a fixed standardized dimension window.

7.2 An exact saddle quotient at every fixed order

Assign weight $j - 2$ to λ_j for $j \geq 3$ and weight j to η_j for $j \geq 1$. For $k \geq 0$, define G_k by summing over finite nonnegative multi-indices $(a_j)_{j \geq 3}$ and $(b_j)_{j \geq 1}$ satisfying

$$\sum_{j \geq 3} (j - 2)a_j + \sum_{j \geq 1} j b_j = 2k.$$

If $d = \sum_{j \geq 3} j a_j + \sum_{j \geq 1} j b_j$, their contribution is

$$(d - 1)!! \prod_{j \geq 3} \frac{(\lambda_j / j!)^{a_j}}{a_j!} \prod_{j \geq 1} \frac{(\eta_j / j!)^{b_j}}{b_j!}. \quad (7.10)$$

The weight condition makes d even. The terms with every $b_j = 0$ give precisely C_k . Define

$$Q_0 = 1, \quad Q_k = G_k - \sum_{i=1}^k C_i Q_{k-i} \quad (k \geq 1). \quad (7.11)$$

Theorem 7.1 (All fixed orders of the binary probability). *For each fixed integer $K \geq 0$,*

$$p_N = e^{h_N(r)} \left(\sum_{k=0}^K Q_k + O_K \left(\frac{L^{4K+4}}{N^{K+1}} \right) \right), \quad (7.12)$$

where $Q_k = O_k(L^{4k} N^{-k})$. In particular,

$$p_N = e^{h_N(r)} \left[1 + \frac{h_N''(r) + h_N'(r)^2}{2A} + \frac{f_N'''(r) h_N'(r)}{2A^2} + O \left(\frac{L^8}{N^2} \right) \right]. \quad (7.13)$$

Proof. Choose $0 < \varepsilon < 1/6$, for example $\varepsilon = 1/12$, and use the standardized window $|z| \leq N^\varepsilon$ about r . Its dimensional width is $N^\varepsilon \sigma = o(r)$. The denominator tails outside this window are $O(e^{-cN^{2\varepsilon}} + e^{-cN})$ relative to the denominator saddle scale, by Section 3. At every feasible integer m , $U(N, m) \leq T(N, m)$, so the same absolute tail bound holds for the numerator. Dividing by $e^{h_N(r)} = e^{-O(L^2)}$ still gives a quantity smaller than every inverse power of N .

This stronger window is essential. A generic estimate e^{-cL^2} from a narrower window does not suffice for the rare-event numerator without comparing its constant to the unknown constant in the penalty.

For the central expansion through weight $2K + 1$, Taylor-expand f_N through derivative order $2K + 3$, and h_N through order $2K + 1$. Their standardized remainders are respectively

$$O_K(N^{-K-1}|z|^{2K+4}), \quad O_K(L^2 N^{-K-1}|z|^{2K+2}).$$

The full correction to $-z^2/2$ is $o(1)$ on the chosen window: the first contributions have orders $N^{-1/2}|z|^3$ and $L^2 N^{-1/2}|z|$. Finite exponential Taylor expansion in $N^{-1/2}$ therefore bounds the integrated remainder by a fixed Gaussian-polynomial integral times $L^{4K+4} N^{-K-1}$. More explicitly, at weight $2K + 2$ there are at most $2K + 2$ penalty factors, since each has weight at least one, and each contributes at most L^2 . Higher remainders are bounded by the same order after integration against a slightly weakened Gaussian. This avoids extra logarithmic powers from a supremum bound on z .

Parity is still weight parity: each phase factor adds degree two beyond its weight, while each penalty factor has equal degree and weight. Thus odd weights have zero Gaussian integral. Lemma 4.1 applies term by term to the finitely many monomials; their bounded powers of L are absorbed by the exponential Poisson error. Extending the polynomial sums outside the window also costs a superalgebraically small amount. The normalized numerator thus has even coefficients G_k and the asserted remainder. Divide by the denominator expansion and use finite series division to obtain (7.11) and (7.12).

At weight two, the pure phase terms C_1 cancel. The remaining coefficient is

$$Q_1 = (\eta_2 + \eta_1^2 + \eta_1 \lambda_3)/2,$$

which is (7.13). □

[write] **Added 5 October 2026, batch 102.** Two steps of this proof are condensed: the bound on “higher remainders” after integration “against a slightly weakened Gaussian”, and the claim that the window $|z| \leq N^\varepsilon$ yields the remainder $L^{4K+4}N^{-K-1}$ without extra logarithms. The intake checked the N^ε tail argument and found it sound; a written-out version of the remainder bound is Question 8. The intake rederived Q_1 by hand; Q_1 and (7.14) are also checked by the source’s symbolic verifier (Section 10), whose receipt the intake reproduced.

For reference, the next exact coefficient produced by the same algorithm is

$$\begin{aligned} Q_2 = & \frac{\eta_1^4}{8} + \frac{5\eta_1^3\lambda_3}{12} + \frac{3\eta_1^2\eta_2}{4} + \frac{5\eta_1^2\lambda_3^2}{8} + \frac{\eta_1^2\lambda_4}{4} + \frac{5\eta_1\eta_2\lambda_3}{4} \\ & + \frac{\eta_1\eta_3}{2} + \frac{5\eta_1\lambda_3^3}{8} + \frac{2\eta_1\lambda_3\lambda_4}{3} + \frac{\eta_1\lambda_5}{8} + \frac{3\eta_2^2}{8} + \frac{5\eta_2\lambda_3^2}{8} \\ & + \frac{\eta_2\lambda_4}{4} + \frac{5\eta_3\lambda_3}{12} + \frac{\eta_4}{8}. \end{aligned} \quad (7.14)$$

This illustrates that the coefficients use exact gamma derivatives and not merely their limiting powers of N .

7.3 The leading penalty and the binary saddle shift

Equations (7.6)–(7.8) turn (7.13) into

$$p_N = e^{H(r)} \left[1 + J(r) + \frac{H''(r) + H'(r)^2}{2A} + \frac{f_N'''(r)H'(r)}{2A^2} + O(L^8/N^2) \right]. \quad (7.15)$$

In particular $p_N \sim e^{H(r)}$ and

$$\frac{N}{L^4} (p_N e^{-H(r)} - 1) \longrightarrow \frac{1}{4}. \quad (7.16)$$

The term $H'(r)^2/(2A)$ is asymptotic to $L^4/(4N)$; the others have smaller logarithmic order. Replacing h_N by $H + J$ in the derivative corrections changes the result by $O(L^8/N^2)$, and expansion of e^J is valid at that same accuracy.

The binary saddle s satisfies $g'_N(s) = 0$. Since $h_N''/A = O(L^2/N) = o(1)$ on I_N , g_N remains strictly concave there, and its derivative keeps the endpoint signs. Thus it has a unique dominant central saddle. Taylor expansion of its equation gives

$$s - r = \frac{h_N'(r)}{A} + O(L^3/N), \quad \frac{s - r}{\log N} \longrightarrow 1. \quad (7.17)$$

This shift is negligible relative to $\sigma \asymp \sqrt{N}/L$, but is significant for correction terms. The one-saddle quotient automatically accounts for it and avoids requiring a separate numerator saddle calculation.

8 The explicit binary correction and the collision interpretation

We now convert the exact-saddle probability into a more accessible carrier using the same u as in (6.2).

Theorem 8.1 (Rational correction). *Let*

$$\mathcal{R}(u) = \frac{u(u+2)(u^4 + 8u^3 + 22u^2 + 20u + 8)}{4(u^2 + 4u + 2)^2}. \quad (8.1)$$

Then

$$p_N = e^{-u^2/2-u} \left[1 + \frac{\mathcal{R}(u)}{N} + O\left(\frac{(\log N)^8}{N^2}\right) \right]. \quad (8.2)$$

Consequently,

$$p_N e^{u^2/2+u} \rightarrow 1, \quad \frac{N}{(\log N)^2} (p_N e^{u^2/2+u} - 1) \rightarrow \frac{1}{4}, \quad \frac{\log p_N}{(\log N)^2} \rightarrow -\frac{1}{2}. \quad (8.3)$$

The binary count has the relative carrier

$$c_N = \sqrt{\frac{2}{P(u)}} \exp\left(\frac{Nu(u+1)}{u+2} - \frac{u}{2} - \frac{u^2}{4}\right) \left(1 + O\left(\frac{(\log N)^4}{N}\right)\right). \quad (8.4)$$

Proof. Use $x_0 = 2N/(u+2)$ and the following rational functions:

$$\begin{aligned} D &= u + 4 + 2/u, & E &= 2u + 6 - 4/u^2, \\ F &= (u+1)(u+2), & G &= (u+2)(3u+4), \\ B_0 &= -F/2 + 1/u. \end{aligned} \quad (8.5)$$

Here B_0 is unrelated to the integral abbreviation $B = N - 1$. We have $-H_N''(x_0) = D/x_0$ and $H_N'''(x_0) = E/x_0^2$.

The differentiated gamma quotient expansion near x_0 gives

$$f_N'(x) = H_N'(x) - \frac{1}{x} - \frac{3q}{x^2} - \frac{2q^2}{x^3} + \frac{1}{2q} + O(L^5/N^2). \quad (8.6)$$

One can obtain this directly by differentiating $q \log M - \log \Gamma(q+1) + q(q-1)/(2M)$ with its differentiated remainder, and using $\psi(q+1) = \log q + 1/(2q) + O(q^{-2})$. The same analytic-disk method as in Section 7, or fixed-order polygamma estimates, bounds the differentiated remainder and avoids differentiating an unspecified error term. At $q_0 = ux_0/2$, (8.6) yields

$$f_N'(x_0) = B_0/x_0 + O(L^5/N^2), \quad r = x_0 + B_0/D + O(L^3/N). \quad (8.7)$$

The last bound follows from one Newton correction and local concavity: the second-order phase term and the variation of the first correction each contribute $O(L^5/N^2)$ to the derivative equation, then division by $A \asymp L^2/N$ gives $O(L^3/N)$.

Put $H_0(x) = -2N(N-x)/x^2$. At $x = x_0$,

$$H_0 = -u(u+2)/2, \quad H'_0 = F/x_0, \quad H''_0 = -G/x_0^2. \quad (8.8)$$

Expansion of the exact penalty $h_N = H + J + O(L^6/N^2)$ at x_0 gives the cancellation

$$h_N(x_0) = -u(u+2)/2 + u/x_0 + O(L^6/N^2). \quad (8.9)$$

For clarity, expand $H = H_0(1 - 1/N)(1 + 1/x_0)^{-1}$ and $J = -x_0(N - x_0)(N - 1)/(2M(x_0)^2)$. Their terms of order x_0^{-1} sum to u/x_0 ; the omitted terms are bounded by $O(L^6/N^2)$. Transport to r using (8.7) adds $FB_0/(Dx_0)$ at the same accuracy.

In the Gaussian correction (7.13), replace A by D/x_0 , f'''_N by E/x_0^2 , and the penalty derivatives by (8.8). All replacement errors are $O(L^8/N^2)$. Thus

$$p_N e^{u(u+2)/2} = 1 + \frac{1}{x_0} \left[u + \frac{FB_0}{D} + \frac{F^2 - G}{2D} + \frac{EF}{2D^2} \right] + O(L^8/N^2). \quad (8.10)$$

Since $B_0 = -F/2 + 1/u$, the two F^2 terms cancel. Rational simplification gives

$$u + \frac{FB_0}{D} + \frac{F^2 - G}{2D} + \frac{EF}{2D^2} = \frac{u(u^4 + 8u^3 + 22u^2 + 20u + 8)}{2P(u)^2}.$$

Finally $1/x_0 = (u+2)/(2N)$, proving (8.2). Before cancellation individual corrections can have order L^4/N ; their squares explain the safe $O(L^8/N^2)$ remainder. Cancellation of the coefficient does not justify shrinking that remainder. Since $\mathcal{R}(u) \sim u^2/4$ and $u \sim L$, (8.3) follows. Multiply (8.2) by (6.4) to obtain (8.4). $\square$

8.1 A Lambert W equivalent

If $w = W_0(\sqrt{N/2})$ and $v = 2w$, then $v^2 e^v = 2N$. Comparing the increasing defining equations gives

$$u = v - 2/v + O(v^{-2}), \quad u^2/2 + u = v^2/2 + v - 2 + O(v^{-1}).$$

For example, subtract their logarithms to get $u + 2 \log u + \log(1 + 2/u) = v + 2 \log v$, then use the mean-value theorem and one substitution. Therefore

$$p_N \sim \exp(2 - 2w^2 - 2w). \quad (8.11)$$

Its relative error is only $O(1/\log N)$, much weaker than (8.2). The factor e^2 cannot be discarded by replacing $u(u+2)$ by u^2 in the implicit equation.

8.2 Why the probability has this exponent

After subtracting the mandatory diagonal ones, a fixed-dimension table is a uniform weak composition of q into M cells. For any particular residual cell X ,

$$\mathbb{P}(X \geq k \mid m) = \frac{(q)_{\underline{k}}}{(M+q-1)_{\underline{k}}}, \quad 0 \leq k \leq q. \quad (8.12)$$

Here $(a)_k = a(a-1)\cdots(a-k+1)$, with $(a)_0 = 1$. The formula follows by subtracting k from that cell and taking the ratio of the two composition counts. Let D_{bad} count diagonal cells with positive residual and C_{bad} count off-diagonal cells with residual at least two. Then

$$\mathbb{E}(D_{\text{bad}} \mid m) = \frac{mq}{M+q-1}, \quad (8.13)$$

$$\mathbb{E}(C_{\text{bad}} \mid m) = \frac{(M-m)q(q-1)}{(M+q-1)(M+q-2)}. \quad (8.14)$$

The first expectation is zero when $q = 0$, and the second is zero when $q < 2$; these conventions replace any degenerate rational expression. At an integer $m = x_0 + O(\log N)$, these means are $u + o(1)$ and $u^2/2 + o(1)$, respectively. Binary means both bad counts vanish, so these expectations explain the two parts of the exponent. They are an interpretation, not a proof of a relative zero-event probability. A total-variation or Gaussian approximation with a merely small absolute error could overwhelm this event. The exact product, strengthened tails and quotient expansion above prove the probability directly; no full collision-distribution theorem is claimed.

Remark 8.2 ([write] How slowly the displayed limits are approached). Added 5 October 2026, batch 102. At the sizes $N = 100, 500, 1000, 2000, 4000, 8000$ the intake compared the exact counts b_N and c_N with their carriers, in 60-digit floating point (not interval arithmetic). The rational correction is accurate: $(p_N e^{u^2/2+u} - 1)/(\mathcal{R}(u)/N)$ equals 0.9984, 0.9889, 0.9895, 0.9911, 0.9929, 0.9946. So are the count carriers: b_N divided by the carrier of (6.4), minus one, is 5.6, 8.9, 7.8, 6.3, 4.8, 3.6 in units of 10^{-3} , and the same quantity for (8.4) is 0.040 at $N = 100$ and 0.0050 at $N = 8000$. The scaled limits in (8.3), however, use $u \sim \log N$, and $u/\log N \rightarrow 1$ only slowly ($u = \log N - 2 \log \log N + \log 2 + o(1)$): at $N = 8000$ the second quantity in (8.3) is 0.145 (limit $1/4$) and the third is -0.284 (limit $-1/2$). The Lambert form (8.11) has relative error $O(1/\log N)$, as stated: the exponent difference $(u^2/2+u) - (v^2/2+v-2)$ behaves like $4/v$ (checked to $N = 10^{1000}$), and $p_N/\exp(2-2w^2-2w)$ is only between 0.50 and 0.63 for the six sizes above. For numerical work, (8.2) with the exact implicit u should be used, not its limits. These are finite uncertified evaluations (compare Remark 2.2).

9 Controlled inversion and integer thresholds

For $X > 1$ define the thresholds

$$\nu_b(X) = \min\{n \geq 0 : b_n \geq X\}, \quad \nu_c(X) = \min\{n \geq 0 : c_n \geq X\}. \quad (9.1)$$

The question is intrinsically discrete. An approximation with error tending to zero can still give the wrong ceiling near an integer, so the final conclusions will be two-sided ceiling brackets.

9.1 Differentiating the exact smooth carrier

Extend the size variable to real y sufficiently large, keeping exactly the gamma phase (2.2). Let r_y denote its unmarked dominant saddle, $A_y = -f_y''(r_y)$, and let $C_\ell(y)$ be the same derivative contractions. For fixed $K \geq 1$, set

$$\begin{aligned} S_K(y) &= e^{f_y(r_y)} \sqrt{2\pi/A_y} V_K(y), \\ V_K(y) &= \sum_{\ell=0}^{K-1} C_\ell(y), \quad F_K(y) = \log S_K(y). \end{aligned} \quad (9.2)$$

Eventually $V_K(y) = 1 + O_K(y^{-1}) > 0$. The count theorem gives, at integer n ,

$$\log b_n = F_K(n) + O_K(n^{-K}). \quad (9.3)$$

We shall *not* differentiate this asymptotic error.

Put $P_y(x) = M(x) + y - x$ and $q = y - x$. At fixed x ,

$$\partial_y f_y = \psi(P_y) - \psi(q + 1), \quad \partial_y \partial_x f_y = (M'(x) - 1)\psi_1(P_y) + \psi_1(q + 1) > 0 \quad (9.4)$$

near the saddle. Here $P_y \asymp x^2$, $P'_y = O(x)$ and $P''_y = 1$. For $j \geq 1$, a typical term in the j th x derivative of $\psi(P_y)$ is a constant multiple of

$$\psi^{(k)}(P_y)(P'_y)^{2k-j}(P''_y)^{j-k} = O(x^{-j}),$$

where the nonnegative exponents describe the usual quadratic-chain derivative formula. The q term is $O(q^{-j})$. Thus

$$\partial_y f_y^{(j)}(x) = O_j(x^{-j}) \quad (9.5)$$

on fixed proportional saddle neighborhoods. The implicit-function theorem and (9.4) give

$$r'_y = \frac{\partial_y \partial_x f_y(r_y)}{A_y} = O(r_y/y) > 0. \quad (9.6)$$

Consequently, direct differentiation of the exact quantities yields

$$A'_y = O(A_y/y), \quad \lambda'_j(y) = O_j(y^{-j/2}), \quad C'_\ell(y) = O_\ell(y^{-\ell-1}). \quad (9.7)$$

For example, the total derivative of $f_y^{(j)}(r_y)$ is $O(r_y^{-j})$ by (9.5), (9.6) and the next x derivative bound. The derivative of each finite contraction then has one extra factor y^{-1} at its weight.

The saddle's x derivative vanishes in differentiating its phase value. Equations (9.4)–(9.7) therefore prove

$$\begin{aligned} F'_K(y) &= \psi(M(r_y) + y - r_y) - \psi(y - r_y + 1) + O_K(y^{-1}) \\ &= \log y - 2 \log \log y + \log 2 + o(1) \sim \log y > 0. \end{aligned} \quad (9.8)$$

This is a derivative statement about the explicit smooth carrier, obtained independently of (9.3).

Theorem 9.1 (Exact carrier inverse bracket). *For every fixed $K \geq 1$ and all sufficiently large X , the equation*

$$F_K(y_K(X)) = \log X \quad (9.9)$$

has a unique sufficiently large solution. It satisfies $y_K(X) \sim \log X / \log \log X$. There exist constants C_K, X_K such that, for $X \geq X_K$,

$$\left[y_K - \frac{C_K y_K^{-K}}{\log y_K} \right] \leq \nu_b(X) \leq \left[y_K + \frac{C_K y_K^{-K}}{\log y_K} \right]. \quad (9.10)$$

Proof. Since $F_K(y) \sim y \log y$ and its derivative is positive eventually, its sufficiently large inverse exists uniquely and has the claimed scale. The sequence b_n is strictly increasing for $n \geq 1$: for each old dimension its stars-and-bars count is nondecreasing as n increases, and the new dimension contributes one extra table.

Let the absolute error in (9.3) be at most $D_K n^{-K}$ eventually. By (9.8), a horizontal shift $\delta = C_K y_K^{-K} / \log y_K$ changes the smooth logarithmic carrier by more than this error when C_K is sufficiently large. At $n = \lceil y_K + \delta \rceil$, monotonicity of the carrier and the lower count bound give $b_n \geq X$. Every integer $n < y_K - \delta$ has $b_n < X$, first within a fixed neighborhood by the same mean-value argument, and then farther away by monotonicity of b_n . This yields (9.10), including the case in which a bracket endpoint is an integer. $\square$

If the two ceilings in (9.10) coincide, their common value is the exact threshold. A sufficient condition is that y_K remain farther than the stated horizontal error from every integer. There is no uniform assertion that $\nu_b(X) = \lceil y_K(X) \rceil$ at all large thresholds.

The constants in this theorem are non-effective in the present proof. A certified finite integer answer requires validated evaluation of the smooth carrier, effective error constants, and possibly exact neighboring counts. Ordinary high-precision floating-point arithmetic alone does not supply that certificate.

9.2 Elementary seeds with vanishing absolute error

Write $\Lambda = \log X$, and let $v > 0$ be the unique solution of

$$v^2(v+1)e^v = 2\Lambda. \quad (9.11)$$

Define

$$\begin{aligned} n_0 &= \frac{\Lambda(v+2)}{v(v+1)}, \\ z_b(X) &= n_0 - \frac{v}{4} - \frac{1}{2} + \frac{\log(P(v)/2)}{2v}, \\ z_c(X) &= n_0 + \frac{v}{4} + \frac{1}{2} + \frac{\log(P(v)/2)}{2v}. \end{aligned} \quad (9.12)$$

Theorem 9.2 (Elementary inverse brackets). *For each fixed $K \geq 1$,*

$$y_K(X) = z_b(X) + O_K(v^3/n_0). \quad (9.13)$$

For some constants C, X_0 and $X \geq X_0$,

$$\left[z_b - Cv^3/n_0 \right] \leq \nu_b(X) \leq \left[z_b + Cv^3/n_0 \right], \quad (9.14)$$

$$\left[z_c - Cv^3/n_0 \right] \leq \nu_c(X) \leq \left[z_c + Cv^3/n_0 \right]. \quad (9.15)$$

The error v^3/n_0 tends to zero. The smooth elementary inverse roots for b and c differ by $v/2 + 1 + o(1)$, and

$$\nu_c(X) - \nu_b(X) = v/2 + 1 + O(1). \quad (9.16)$$

Proof. For real y , let $u = u(y)$ solve $u(u+2)e^u = 2y$. Define

$$F_0(y) = \frac{yu(u+1)}{u+2}, \quad g_{\pm}(u) = \pm u^2/4 \pm u/2 - \frac{1}{2} \log(P(u)/2).$$

Differentiating the implicit equation gives the exact identities

$$u'(y) = \frac{u(u+2)}{yP(u)} = O(y^{-1}), \quad F'_0(y) = u(y). \quad (9.17)$$

Equation (9.11) is precisely $F_0(n_0) = \Lambda$, with $u(n_0) = v$. The smooth derivation of Theorem 6.1, applied to the exact finite carrier in (9.2), gives

$$F_K(y) = F_0(y) + g_+(u(y)) + O_K(u(y)^4/y). \quad (9.18)$$

Now $z_b = n_0 - g_+(v)/v$ and $z_b - n_0 = O(v)$. Taylor's theorem, $F_0''(y) = O(1/y)$ and $(g_+ \circ u)' = O(u/y)$ show that

$$F_K(z_b) - \Lambda = O_K(v^4/n_0).$$

Division by (9.8), which is asymptotic to v here, proves (9.13). Combining this with the count error, or applying the same mean-value comparison directly, proves (9.14).

For the binary count, the explicit smooth logarithmic carrier is $F_-(y) = F_0(y) + g_-(u(y))$. Its derivative is $u + O(u/y) > 0$. The count formula (8.4) gives $\log c_n = F_-(n) + O(u(n)^4/n)$, and the same Taylor calculation shows its root is $z_c + O(v^3/n_0)$. The needed monotonicity of c_n follows by an injection: append a final row and column with new diagonal entry one and all new off-diagonal entries zero. Deleting that row and column recovers the original table. This raises the size by one while preserving the binary class. Individual fixed-dimension binomial terms need not be monotone. The same ceiling argument proves (9.15).

Finally $z_c - z_b = v/2 + 1$ exactly, while the continuous root errors vanish. Passing to integer thresholds introduces bounded rounding uncertainty, proving (9.16). $\square$

[write] **Added 5 October 2026, batch 102: reading (9.16).** As printed, the “+1” in (9.16) is absorbed by the $O(1)$: the statement is $\nu_c(X) - \nu_b(X) = v/2 + O(1)$. This is a point of wording, not a correction of substance. The exact statement behind it is the identity $z_c - z_b = v/2 + 1$ of the smooth seeds (9.12), and the brackets give an explicit window around it: with $\delta = Cv^3/n_0$ as in (9.14)–(9.15),

$$\frac{v}{2} - 2\delta < \nu_c(X) - \nu_b(X) < \frac{v}{2} + 2 + 2\delta \quad (X \geq X_0),$$

an interval of length $2 + 4\delta \rightarrow 2$ centred at $v/2 + 1$.

Proof [write]. For real a and integer ν , $\lceil a \rceil \leq \nu$ is equivalent to $a \leq \nu$, and $\nu \leq \lceil a \rceil$ implies $\nu < a + 1$. Hence (9.14) gives $z_b - \delta \leq \nu_b < z_b + \delta + 1$, and (9.15) gives $z_c - \delta \leq \nu_c < z_c + \delta + 1$. Subtracting, $z_c - z_b - 1 - 2\delta < \nu_c - \nu_b < z_c - z_b + 1 + 2\delta$, and $z_c - z_b = v/2 + 1$ by (9.12). $\square$

The step (9.18) of the proof above, the real-size form of Theorem 6.1 for the smooth carrier F_K , is asserted as “the smooth derivation” of that theorem, and the derivative bounds (9.7) are argued in one sentence. A written-out version is Question 7.

9.3 All fixed logarithmic orders

Let $w = \log \Lambda$. The equation for v becomes

$$v + 3 \log v + \log(1 + 1/v) = w + \log 2. \quad (9.19)$$

It first gives $v = w - 3 \log w + \log 2 + O(\log w/w)$, whence for either threshold

$$\nu_\bullet(X) = \frac{\Lambda}{w} + (3 \log w + 1 - \log 2) \frac{\Lambda}{w^2} + O\left(\frac{\Lambda(\log w)^2}{w^3}\right). \quad (9.20)$$

The additive seed corrections $O(w)$ and integer errors $O(1)$ are smaller than this remainder.

Every fixed logarithmic order can be generated recursively. Substitute

$$v = w - 3 \log w + \log 2 + \sum_{j=1}^J \frac{P_j(\log w)}{w^j} \quad (9.21)$$

into (9.19) and cancel successive inverse powers of w . At each stage the new polynomial P_j has coefficient one, so the recursion determines it uniquely. For example,

$$P_1(s) = 9s - 3 \log 2 - 1.$$

The residual after P_J is $O_J((\log w)^{J+1}/w^{J+1})$. Since the derivative of the left side of (9.19) tends to one, the error in v has this same order. Substitution into $n_0 = \Lambda(v+2)/(v(v+1))$, whose derivative in v is $O(\Lambda/w^2)$, gives error

$$O_J\left(\frac{\Lambda(\log w)^{J+1}}{w^{J+3}}\right). \quad (9.22)$$

For each fixed J , the omitted $O(w)$ elementary seed correction is negligible compared with (9.22). These are controlled scale expansions; they are not exact rounding prescriptions.

For practical refinement, z_b is a useful starting value for a safeguarded Newton or bisection solver for (9.9). The positivity in (9.8) supplies the local monotonicity. Evaluation error and asymptotic count error must be tracked separately. A finite check illustrates the warning: at $X = b_{1000}$, the elementary seed is approximately 1000.0014164605, so taking its ceiling gives 1001, while the exact threshold is 1000. Even a much more accurate smooth root can lie on the wrong side of the same integer. The two-ceiling formulation is indispensable.

10 Reproducible checks and their precise scope

The accompanying package contains the complete TeX source, four deterministic computational receipts, the checking programs, an immutable-source builder and an actual-archive replay verifier. The programs use Python 3.11 or later, mpmath 1.3.0 and SymPy 1.14.0. Their roles are finite validation and reproducibility; the preceding analytic arguments establish the theorems.

10.1 Exact and symbolic checks

The exact verifier independently computes the binomial sums and the coefficients obtained by multiplying individual cell generating polynomials. Its default range is $0 \leq N \leq 40$. It checks all 820 positive-size triangle entries in that range, 820 conditional binary products, 820 fixed-dimension binomial-transform identities, 1640 residual-cell tail identities, 1640 direct bad-cell expectation comparisons, and the 40 total transforms. The expectation check uses a separate cell-polynomial recurrence carrying summed indicators and includes the zero-residual cases. Source-linked finite fixtures check 26 ordinary and 25 binary terms with the current shifts in (1.10). The code also calculates exact sums at $N = 100, 200, 500, 1000, 2000$.

The symbolic verifier independently enumerates weighted multi-indices, applies exact Gaussian moments, and compares the results with C_1, C_2, Q_1, Q_2 . It generates C_3 as well. A second weighted-exponential route checks the numerator contractions through weight four. Further exact algebra checks the moment formulas, the constants $1/24$ and $1/1152$, the rational identity in Theorem 8.1, $F'_0 = u$, the inverse equation and the seed separation. No coefficient is fitted to numerical data.

Large exact integers are hashed using their unsigned big-endian byte encodings; they are not converted to unbounded decimal strings. Each Python process explicitly sets a 640-digit decimal conversion cap and disables bytecode writes. Every child invocation repeats these settings. Guards use explicit exceptions, so optimized Python cannot remove validation.

10.2 Finite numerical diagnostics

The default numerical cases use 90 decimal digits at unmarked sizes 100, 200, 500, 1000, 2000, together with marks $t = \pm 0.25$ at $N = 500$. The exact gamma phase is retained. A bounded finite-precision bisection solves the saddle equation, checking endpoint signs and residuals. The script compares exact integer sums with the saddle carriers and moment corrections, records the full-support leading local-limit error and the tail mass outside I_N , and evaluates both binary carriers and inverse seeds.

At $N = 1000, t = 0$, selected values are

Quantity	Numerical value
Saddle r	315.5139205982
Exact weighted mean	315.6063993121
Exact weighted variance	36.5565672868
Binary probability p_N	$1.3110712105 \times 10^{-6}$

Here “exact weighted” describes the underlying finite integer sums; the displayed evaluations are rounded floating-point values. At $N = 1000$, the residual

$$\frac{b_N}{e^{f_N(r)} \sqrt{2\pi/A}} - \sum_{\ell=0}^{K-1} C_\ell$$

is approximately -1.1264×10^{-5} for $K = 1$, 6.0743×10^{-10} for $K = 2$, and -5.5610×10^{-13} for $K = 3$. These numbers are useful consistency checks; a short finite range need not yet show the limiting sign of a coefficient.

The inverse diagnostic at $X = b_{1000}$ also computes the $K = 2$ smooth-carrier root, approximately 1000.0000000001412, using a bounded sign-bracketed secant iteration. Its computed log residual is about -1.20×10^{-44} . This is not an interval enclosure and does not supply the asymptotic constant in (9.10). Its position just above the exact integer reinforces the rounding warning rather than certifying a ceiling rule.

10.3 Build and replay discipline

The public source inventory is complete: every file other than the manifest itself is hashed, and extra files, directories, symlinks or malformed names are rejected. The builder does not regenerate its own manifest. It requires a new output directory outside the source tree and compares source inventories before and after its work.

The PDF is compiled from a disposable copy with a private TeX format and cache, three passes, fixed date and metadata, and shell escape disabled. Overfull boxes, missing glyphs and unresolved references are rejected. The builder requires the regenerated PDF and all four computational receipts to match their frozen originals byte for byte. It writes a deterministic archive with fixed member order, timestamps and modes.

The archive replay first validates the actual input ZIP against the trusted source package. It extracts two separate copies and builds under ordinary Python and Python with optimization. It compares all seven top-level output files: the PDF, ZIP, four receipts and build-check record. It also compares archive member bytes and metadata, requires each regenerated ZIP to equal the original input archive, and checks that the trusted source tree, both extracted source trees and input archive remain unchanged. The 241 finite guard cases and operational bounds are enumerated in the computation note. These filesystem tests are not a general security proof against hostile concurrent changes.

The document was rendered and visually inspected page by page before packaging. Reproducibility is asserted for the recorded toolchain, not arbitrary TeX or library versions. All finite checks remain distinct from effective asymptotic bounds, external peer review and formal proof verification.

[write] **Added 5 October 2026, batch 102: shipped files and intake reruns.** In ProveIt the package is filed in the report's own layout: the checking programs and the builder in `code/` (the builder `build.py` was at the package root), the four receipts, the OEIS fixture `source_prefixes.json` and `requirements.txt` in `data/`, and the two notes `SOURCES.md` and `COMPUTATION.md` at the report root. The delivered PDF and manifest are not shipped. The programs locate their inputs relative to the delivered layout, so nothing runs in place; the report's README gives rerun routes on copies. At placement the intake verified the manifest (30 of 30 entries) and reran `exact_counts.py`, `algebra_checks.py` and `saddle_diagnostics.py` on a copy of the delivered layout; at the write it reran all three on a copy of the shipped files with the fixture restored next to the programs. Each time all three receipts were byte-identical to the delivered ones (Python 3.14.4, mpmath 1.3.0, SymPy 1.14.0, Windows). The guard suite was not run: it requires a POSIX `/tmp` and rejects Windows paths; its delivered receipt records 241 passing cases. The builder's full build fails on Windows at its first receipt comparison, because child output captured in text mode acquires CRLF line endings (the receipt is identical after stripping them), and its PDF comparison requires the recorded TeX Live toolchain; the replay verifier depends on the builder. The committed `article.pdf` is a MiKTeX build of the written text, not the delivered PDF.

11 Limits and further questions

The exact formulas (1.3)–(1.8), the ascent-sequence/table bridge and the historical rigid subclass belong to the cited enumerative literature. The contribution developed here is the relative saddle analysis with its explicit error organization and consequences for this specific model. The result is self-contained modulo standard gamma asymptotics and Gaussian Poisson summation, whose needed uniform estimates have been explained.

Several useful questions remain separate from the results proved here.

1. **Effective error constants.** Explicit uniform gamma bounds and validated saddle evaluation could turn the eventual inverse brackets into finite-size certificates. Our finite diagnostics do not accomplish this.
2. **Growing expansion order.** The coefficient algorithms determine every fixed order. Their growth as the order increases, optimal truncation and any exponentially improved remainder require additional analysis.
3. **Complex dimension marks.** Bounded real marks suffice for the dimension laws and moments established here. Uniform complex marks or a global analytic generating-function theorem are not asserted.
4. **Further statistics.** The expectations in (8.13)–(8.14) suggest studying the joint collision

structure. A full distribution theorem, particularly one strong enough for relative rare probabilities, is not a consequence of those expectations.

5. **Other difference parameters.** The $d = 0$ and $d = 1$ exact models admit the present analysis. General fixed positive d , and the stabilized $d \rightarrow \infty$ family, must be treated from their own enumeration and cannot be inferred by changing the signs in our formulas.

The asymptotic conclusions do not establish worldwide priority, absence of equivalent results in uninspected sources, external peer review or machine-checked proofs. The accompanying source and computation notes delimit what was actually checked. In particular, source text access is not represented as inspection of unavailable page images, and finite numerical agreement is not represented as a theorem about an infinite range.

11.1 Further questions and research

[write] **Added 5 October 2026, batch 102.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. Questions 1–5 are the five items above; Questions 6–9 are steps that the text asserts or condenses; Question 10 concerns the external inputs. The intake found no false claim in the source; the one point of wording, in (9.16), is explained in the note after Theorem 9.2.

1. **Effective constants and certified thresholds** (source, item 1 above). Every O -constant and threshold in Theorems 2.1–9.2 is an existence constant (Remark 4.2). Explicit uniform gamma and polygamma bounds in Lemma 3.1, (3.9) and Lemma 4.1, with validated evaluation of the saddle, would turn the brackets (9.10), (9.14) and (9.15) into finite certificates. The finite evidence shows why this matters: C_1 changes sign only near $N \approx 1.59 \times 10^4$ (Remark 2.2), the displayed limits are approached slowly (Remark 8.2), and the elementary seed rounds the wrong way at $X = b_{1000}$ (Section 9). An interval computation of (2.10) would also make the sign change of Remark 2.2 a theorem.
2. **Growing expansion order** (source, item 2). How do C_ℓ and Q_k grow with the order; is there an optimal truncation, and an exponentially improved remainder? Missing: any estimate uniform in the order. The source’s data at $N = 1000$ (Section 10): the residuals after $K = 1, 2, 3$ terms are about -1.13×10^{-5} , 6.07×10^{-10} and -5.56×10^{-13} , gains of factors about 1.9×10^4 and 1.1×10^3 .
3. **Complex dimension marks** (source, item 3). Theorem 2.1 is uniform for bounded real t only. Uniform complex marks, or a global analytic statement about $B(z, h)$, would give, for example, large-deviation rates and analytic continuation in the mark; nothing of this is asserted.
4. **The joint collision structure** (source, item 4, and Section 8.2). The expectations (8.13)–(8.14), asymptotically u and $u^2/2$, “explain” the exponent of $p_N \sim e^{-u^2/2-u}$, as if $(D_{\text{bad}}, C_{\text{bad}})$ were nearly independent Poisson variables. The source states this as an interpretation only. Missing: a joint distribution theorem for the two bad counts that is strong enough to give the relative probability of the rare event $\{D_{\text{bad}} = C_{\text{bad}} = 0\}$ (a total-variation statement is not).
5. **Other difference parameters** (source, item 5). Only $d = 0$ and $d = 1$ of [5] are treated. Fixed $d \geq 2$ and the stabilized $d \rightarrow \infty$ family need their own exact enumeration and analysis; changing signs in the present formulas proves nothing.

6. **Higher moments and local corrections** (source, the paragraph after Proposition 5.2). The source asserts that “all higher fixed moments and local corrections are obtainable from the weighted-contraction rule and finite series division”. Sketch: repeat Section 4 with a factor z^k , which the polynomial tail bound after (3.14) makes uniformly integrable. Missing: the general statement with its remainder (only $k = 1, 2$ are proved, in Proposition 5.2) and the local expansion (5.1) beyond its first correction.
7. **The smooth elementary carrier** (source, proof of Theorem 9.2). The real-size estimate (9.18), $F_K(y) = F_0(y) + g_+(u(y)) + O_K(u^4/y)$, is asserted as “the smooth derivation” of Theorem 6.1, and the bound $C'_\ell(y) = O_\ell(y^{-\ell-1})$ in (9.7) is argued in one sentence. Sketch: the proof of Theorem 6.1 uses only the exact phase at real arguments, and each contraction is a polynomial in the λ_j , whose y -derivatives (9.7) follow from (9.5) and (9.6). Missing: the written argument that every remainder in the proof of Theorem 6.1 is uniform for real y , and the chain-rule bookkeeping for the contractions.
8. **The remainder in Theorem 7.1** (source, its proof). The statement that “higher remainders are bounded by the same order after integration against a slightly weakened Gaussian” is a sketch. Missing: the explicit domination of the Taylor remainders of f_N and h_N on the window $|z| \leq N^\varepsilon$ that yields $L^{4K+4}N^{-K-1}$ without additional logarithmic factors. The intake checked the window and tail argument and believes the bound.
9. **Moderate deviations** (source, after Proposition 5.2): the relative Gaussian approximation for $|z| = o(N^{1/6})$, argued in two sentences. *Resolved at the write*: Proposition 5.3 proves it, with the error $\exp(O(|z|^3 N^{-1/2} + N^{-1}))$. Listed for the record.
10. **External inputs and priority** (source, Section 1 and SOURCES.md). The source’s search was bounded, and its page-level readings of [1, 2, 3, 4, 5, 6, 7, 8, 9] were not checked by ProveIt. Open: whether the relative results here, in particular the answer to A098568’s question, appear in sources the search did not reach, and whether Kotešovec’s finer logarithmic expressions [10] have a published proof (the inspected records cite none).

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- [10] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, [A098569](#), [A098568](#) and [A121690](#); records checked 5 October 2026. See the source note for revisions, offset conventions and the prior logarithmic postings by Václav Kotešovec.
- [11] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, [Section 5.11](#) (gamma and digamma asymptotics) and [Section 5.15](#) (polygamma functions), checked 5 October 2026.